

AI Data Analyst

Project Report

Full-Stack CSV Data Analysis with AI

Abstract

AI Data Analyst is a full-stack web application that simplifies CSV data exploration. Users can upload a dataset, view statistics, inspect records, generate charts, and ask natural-language questions through an AI assistant.

1. Introduction

Data analysis commonly requires repeated tasks such as checking dataset structure, calculating descriptive statistics, creating visualizations, and interpreting results. This project combines these tasks into one browser-based workflow.

2. Problem Statement

Many users have CSV data but lack the time or programming knowledge needed to perform a complete exploratory analysis. The project addresses this by providing an accessible interface for data exploration and AI-assisted interpretation.

3. Objectives

- Provide a simple CSV upload workflow.
- Display dataset statistics and structure.
- Provide a readable data preview.
- Generate useful visualizations.
- Allow natural-language questions about uploaded data.
- Support both laptop and mobile browsers.
- Deploy the complete application publicly.

4. Proposed Solution

The application accepts a CSV upload, processes the dataset through a FastAPI backend, and presents summary statistics, data previews, charts, and AI answers through a React frontend.

5. Key Features

- CSV upload and dataset management.
- Overview and descriptive statistics.
- Data preview table.
- Data visualizations.
- Natural-language Ask AI interface.
- Responsive desktop and mobile interface.
- PostgreSQL integration.
- Google Gemini integration.
- Public cloud deployment using Render.

6. Technology Stack

Technology	Role
React + Vite	Interactive frontend and build tooling
FastAPI	REST API backend
Python + Pandas	Data processing and analysis
PostgreSQL	Persistent database storage
Google Gemini	AI-assisted natural-language analysis
Git + GitHub	Version control and source hosting
Render	Cloud deployment

7. System Architecture

The browser hosts the React frontend. The frontend communicates with the FastAPI backend. The backend processes CSV data, uses PostgreSQL for persistence, and integrates with Google Gemini for AI-assisted questions.

Architecture: User → React/Vite Frontend → FastAPI Backend → Pandas / PostgreSQL / Gemini

8. Functional Workflow

1. Open the deployed application.
2. Upload a CSV file.
3. Select the uploaded dataset.
4. Review Overview statistics.
5. Inspect the Data tab.
6. Explore Charts.
7. Ask a natural-language question in Ask AI.
8. Review the generated response.

9. Results

The completed application provides an end-to-end workflow from CSV upload to data exploration and AI-assisted questions. The frontend and backend are deployed publicly and the interface has been adapted for laptop and mobile browser use.

10. Testing

Test	Expected Result	Status
CSV upload	Dataset uploads successfully	Completed
Data preview	Records are displayed	Completed
Overview	Statistics are displayed	Completed

Charts	Visualizations render	Completed
Ask AI	AI response is returned	Completed
Dataset deletion	Dataset can be removed	Completed
Mobile interface	Core UI works on mobile	Completed
Deployment	Public services are accessible	Completed

11. Challenges and Solutions

- Backend deployment dependency issue — required PostgreSQL driver/dependencies were included for production deployment.
- Frontend/backend production connection — the frontend was configured to communicate with the deployed FastAPI backend.
- Mobile upload experience — responsive styling and the file-upload interaction were adjusted for smaller screens.
- Desktop-only empty-state wording — the message was changed to reference the upload control above.
- Free hosting cold starts — the deployment accounts for possible first-request delays after inactivity.

12. Deployment

Frontend: <https://ai-data-analyst-frontend-1wyn.onrender.com>

Backend: <https://ai-data-analyst-backend-hoxe.onrender.com>

API documentation: <https://ai-data-analyst-backend-hoxe.onrender.com/docs>

13. Future Scope

- Support Excel and additional data formats.
- Generate downloadable PDF/Excel analysis reports.
- Add advanced statistical analysis and filtering.
- Add authentication and private user workspaces.
- Add saved dashboards and analysis history.
- Add a custom domain and stronger production monitoring.

14. Conclusion

AI Data Analyst demonstrates a practical combination of web development, data processing, database storage, visualization, and generative AI. The completed system provides a simple workflow for turning CSV data into understandable information and AI-assisted insights.

15. Project Links

GitHub: <https://github.com/Siriyala-Nishar-10/AI-Data-Analyst>

Live Demo: <https://ai-data-analyst-frontend-1wyn.onrender.com>